FEAT-ZD-48600 · BE-03 Database changes · Completed database changes

Backend agent · item task.db_changes · source hobasa-backend-agent/outputs/FEAT-ZD-48600/task.db_changes.md · approved by Developer sim.developer

task.db_changes: FEAT-ZD-48600

Database-change check for the Backend lane of FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"). This is step BE-03 Database changes. It covers the Backend tasks TSK-05 (STORY-05) and TSK-06 (STORY-06) in ap_aging.

Result: this feature needs no database changes, and none are outstanding. The step outcome ("Required database changes are completed") holds because there are no required changes.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	BE-03 Database changes	record:brief
Owner / gate	Developer / no gate (automated checks, the independent judge and sampled human evaluation)	record:brief
Inputs	feat.tasks (b0fd4d2), feat.design (722a98f)	record:brief
Repository	ap_aging	tool:repo-access; feat.tasks (section 1)
Backend tasks in scope	TSK-05 `overdue_with_risk`, TSK-06 `overdue_rows_to_json`	feat.tasks (section 1, TSK-05, TSK-06)

1. What the design requires of storage

Check	Finding	Source
Storage in the design boundary	"Storage, database, schema migration" is listed as **out of this design**	feat.design (A1, Data row)
New data in this feature	One new record key `paid_date`, one new reject reason `'paid_date'`, and in-memory delay history, risk map, overdue rows and CT-1 JSON	feat.design (A1, Data row)
Technology	No new repository, service, runtime, library or storage; Python 3.9 standard library and Node.js 22 CommonJS only (DD-1)	feat.design (A7, DD-1)
Purity of new functions	Every new function is pure and in-memory: no I/O, no network, no mutation of inputs (NF-10)	feat.design (A5, NF-10)
What reopens the platform	A storage or network call from the libraries is one of the triggers that would reopen DES-01; this design triggers none of them	feat.design (A8)
ap_aging data structures	Input invoice (DC-1, a plain dict), overdue row (a dict), CT-1 payload (a JSON string)	feat.design (A3 DC-1, B3.3)
ap_aging module changes	`src/ap_aging/aging.py` (constants, `overdue_with_risk`, `overdue_rows_to_json`, `import json`, `List` typing) and `src/ap_aging/__init__.py` (exports, version step)	feat.design (B3.1); feat.tasks (TSK-05, TSK-06 "Files in budget")
CT-1 transport	No transport layer is added; CT-1 is a serialised string that the consumer parses (DD-3)	feat.design (A3, DD-3); feat.tasks (TSK-06 "Must not change")

2. What the repository holds today

Check	Finding	Source
Database in ap_aging	ap_aging has no database; no migrations exist or are required	tool:repo-access
Imports in the aging module	`aging.py` imports only `date` from `datetime`, `Decimal` from `decimal`, and `Dict` and `Iterable` from `typing`	file:ap_aging/src/ap_aging/aging.py
Existing function shape	`aging_buckets(invoices, as_of)` loops over an iterable of invoice dicts and returns a dict of totals per bucket	file:ap_aging/src/ap_aging/aging.py
Language and stack	Python 3.9 (standard library)	file:ap_aging/CLAUDE.md

3. Required database changes

Change	Required?	Status	Source
Schema, table, column, index or migration	No: the design keeps storage out of scope, and ap_aging has no database	Nothing to do	feat.design (A1, DD-1); tool:repo-access
Data backfill or reference data	No: the new ap_aging data is in-memory rows and a JSON string built by pure functions	Nothing to do	feat.design (A1, NF-10, B3.3)
DBA or infrastructure prerequisite	No: there is no database change to prepare	Nothing to request	feat.design (A1); tool:repo-access

The new `paid_date` key is a field of the in-memory invoice record (DC-1), built by `transform.to_model` in ap_pipeline under TSK-01 (Data Engineering lane). In ap_aging, extra keys such as `paid_date` are ignored (feat.design A2 HC-1, A3 DC-1, B3.3; feat.tasks TSK-01).

4. Effect on the next steps

- The Backend tasks have no database dependency. TSK-05 depends on nothing in code, and TSK-06 depends on TSK-05 (feat.tasks section 1, TSK-05, TSK-06).
- Agent judgement: if a later change added storage, or a transport between ap_aging and ap_portal, it would reopen DES-01 (feat.design A8), and the database check should then be repeated.

5. Open questions

None from this step. DQ-1 (which production component calls the stages in sequence and carries CT-1) stays open, as carried in feat.design D3. Its interim answer adds no transport (DD-3), so it creates no database change for this feature.

6. Validation check

Check	Result	Source
Storage needs read from the design inputs	Pass: feat.design A1, A3, A5, A7, A8, B3 and feat.tasks TSK-05, TSK-06	feat.design; feat.tasks
Database in the repository	None	tool:repo-access
Conflict with an approved output	None	feat.design; feat.tasks
Blocking questions	None	—

Tool results (recorded by the spine from the real tools)

[Tool result] [ap_aging] No database in this repository's architecture; no migrations required